

Kabir Singh

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ACADEMIC EDUCATION

- Albert Dorman Honors College (ADHC)** and Jordan Hu College of Sciences and Liberal Arts (JHCSLA), New Jersey Institute of Technology, Newark, New Jersey, USA Sep '23 – May '27 (Expected)
- Bachelor of Science; Major – **Mathematical Biology**
 - Minors – **Biomedical Engineering in Technology, Philosophy, History**
 - ADHC Honors Tracks – **Medical Humanities, Global Studies**
 - *Academic Excellence, Honors Merit, Honors Residential Scholarship Recipient (Tuition + Board)*
- Faculty of Engineering, Lunds Tekniska Högskola, **Lunds Universitet**, Lund, Sweden Jan '25 – Jun '25
- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3)

RESEARCH

- Undergraduate Researcher, Mathematical Neuroscience. PI: Dr. Horacio Rotstein, NJIT Aug '26 – Present
- Deriving reduced mathematical descriptions of biophysical neuronal models to characterize their dynamic properties
- Undergraduate Researcher, **History of Medicine and Bioethics**. PI: Dr. Stephen Pemberton, NJIT Aug '26 – Present
- Synthesizing legal, clinical, and cultural developments in American end-of-life care from the late twentieth century to the present
 - Developing original scholarship for submission to a peer-reviewed humanities journal
- Undergraduate Researcher, Computational Bionanotechnology. PI: Dr. Stefano Angioletti-Uberti, Imperial College London Jun '26 – Present
- **BAGEL: Biomolecular Algorithm for Guidance in Energy Landscape**
 - Computational modelling of soft-matter and biomaterial systems
 - Applying statistical mechanics, molecular dynamics, and machine-learning methods to problems in enzyme design and broader protein design
- Undergraduate Researcher, **Computational Biology Team Lead** at KumarLab. PI: Dr. Vivek Kumar, University of Houston (Prev. NJIT) Oct '23 – Present
- Lead undergraduate and graduate students on designing and developing peptide-based therapeutics
 - Started at NJIT and continued computational work with lab relocation to Houston
 - Taught 15+ associates in lab
 - Received > \$15,000 in direct funding for projects
- Undergraduate Researcher, **Mathematical Modeling**. PI: Dr. James Maclaurin, NJIT Oct '25 – May '26
- Applying Hawkes self-exciting point process to model secondary nucleation dynamics in amyloid formation
- Undergraduate Researcher, **Digital History and Urban Studies**. PI: Dr. Louis Hamilton, NJIT Jan '24 – May '24, Jan '26 – May '26
- Digital mapping of how religious institutions shaped urban form in medieval and early modern Italy; field research conducted in Rome and Florence
 - *Power vs. the People* (HIST 328; 18 sites): layered Counter-Reformation Marian shrines against Julius II's 1508 papal axis on the Via Giulia
 - *Processions and Power* (HIST 329; 22 sites): traced how Florence's expansion turned liturgical processions into contested displays of ecclesiastical and household prestige

FUNDING

- Undergraduate Research Opportunities Programme (UROP), Imperial College London Jun '26 – Jul '26
- Competitive research bursary awarded through the Department of Materials, Faculty of Engineering
 - Project: Using protein language models to mimic natural ligands. Supervisor: Prof. Stefano Angioletti-Uberti
 - Personal compensation included

NSF I-Corps Regional Program, Northeast Hub	May '26 – Jun '26
– Co-Founder & Technical Lead, Axioma Therapeutics – Selected into competitive NSF cohort; led customer discovery and lean-startup validation for cyclic peptide therapeutics targeting degenerative eye diseases – Project funding awarded to support customer-discovery activities	
Grace Hopper AI Research Institute, Computational Protein Design	May '25 – Aug '25
– NJIT grant to develop a course on computational protein design – Teaching Assistant	
The Honors College Moonshot Grants for Engaged Scholarship	Jun '25 – Jul '25
– Endowed award to develop interdisciplinary projects that address important social and cultural questions – Designing Male Contraceptive – Award providing labs with funds to support undergraduate-led projects – Personal compensation included. One recipient per annum	
Male Contraceptive Initiative (MCI)	Jan '25 – Jun '25
– Award by the Youth Advisory Board providing funding to support grantee-led project – Designing Male Contraceptive – Personal compensation included	
NJIT Undergraduate Research and Innovation (URI)	Oct '24 – May '25
– Student Seed Grant award for a Phase-2 longitudinal project on application-based research funding lab to support undergraduate-led project – Developing cyclic peptide therapeutic for Multiple Sclerosis	
Travel Award – National Contraception Meeting	May '25
– Awarded by the Male Contraceptive Initiative for travel to Durham, NC, supporting a conference presentation	
NJIT Honors Summer Research Institute (HSRI)	Jun '24 – Jul '24
– Grant award by ADHC and URI providing funds to support undergraduate-led projects – A combination of Multiple Sclerosis project and Male Contraceptive project – Personal compensation included	

PUBLICATIONS

Peer-Reviewed

- C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis. *European Journal of Medicinal Chemistry*, 119173 (2026). doi:10.1016/j.ejmech.2026.119173

Manuscripts in Preparation

- **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. Machine learning for rational design of a ***** targeting cyclic peptide-based male contraceptive. Publication [in preparation], **Patent application** [in preparation], 2026
- J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. Academic Publishing as a Launchpad: How Publishing Research in an Undergraduate Journal Shapes Future Careers. Publication [in preparation], 2026

PRESENTATIONS

- M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor.
UH (Undergraduate Research Day) Poster + Podium 2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. Development of a novel male contraceptive using cyclic peptides to inhibit fertilization.

AAP/ASCI/APSA Joint Meeting	Poster	2026
NJIT AI Exploration Day	Poster	2026
NJIT ADHC (Honors Interdisciplinary Research Forum) First Place	Poster	Fall 2025
NCM (National Contraception Meeting)	Poster	2025
NJIT ADHC (Honors Interdisciplinary Research Forum)	Poster	Spring 2025
NJIT (Dana Knox Research Showcase)	Poster	2025
NJIT (Undergraduate Research and Innovation)	Abstract + Podium	2025
BMES (Biomedical Engineering Society) [Co-Presenter]	Abstract + Poster	2024
NJIT (Undergraduate Research and Innovation)	Abstract + Podium	2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. Mitigating Knowledge Drain: A Case Study in Optimizing Workflow and Training in a High-Turnover, International Peer-Reviewed Undergraduate Science Academic Journal.

NCUR	Poster	2026
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ACADEMIC SERVICE

22nd Editor-in-Chief, Associate Editor (Former), Journal of Young Investigators Dec '24 – Present

- Manage timely publication of high-quality scientific content
- Develop initiatives to further the journal's mission
- Former specialization in Biology, Physical Sciences, and Social Science Departments

Research Chair, American Association of Physicians of Indian Origin (AAPI) PreMed Board Jun '25 – Present

- Lead committee and organize events to achieve research-related objectives on a national level to support students, researchers, physicians, and more of an Indian origin

Board Member, MCI Youth Advisory Board (YAB) Aug '25 – Present

Member, Pre-Medical Ad Hoc Committee, APSA Mar '26 – Present

Member, Goldwater Ambassadors Program Aug '25 – Present

- Selected into competitive process as a mentee, mentored by Goldwater Scholars
- Program specially designed for students from underrepresented or underserved backgrounds

Member, ARTISAN Program Apr '25 – Present

- Inducted in the NSF ART Integrated Student Ambassador Network (ARTISAN) program, based off commitment to translational research in my work

Member, Inclusive Excellence Student Advisory Board Feb '25 – Present

Member, Multifaith Council Jan '25 – Present

Member, Student Conduct Board Dec '24 – Present

- Serve as adjudicator on academic and behavioral cases in student conduct hearings, along with other members of the board, including faculty and NJIT administration

Chair, Chair-in-Training (Former), Highlander Integrity Council Sep '24 – Present

Organizer, 'It's Different for Girls' Documentary Screening and Panel, NJIT ADHC and CSLA Mar '26

Co-Sponsored Colloquium

NJIT Representative, BioNJ 33rd Annual Dinner Meeting & Innovation Celebration Feb '26

Grant Reviewer, MCI YAB Undergraduate Project Grant Dec '25 – Jan '26

Teaching Assistant, Honors First Year Seminar Sep '24 – Dec '24,
Sep '25 – Dec '25

Invited Speaker, NJIT First Aid Squad Sep '25, Oct '25

- Presented to NJIT Board of Trustees as well as the President's Town Hall with squad leadership at the request of NJIT President Teik Lim

Steering Committee Lead, ADHC First Year Retreat Mar '25 – Aug '25

Teaching Assistant, Grace Hopper AI Research Institute May '25 – Aug '25

– Computational Protein Design

Featured Speaker, ADHC Advising Day

Jun '24 & '25

Advisor, President (Former), Secretary (Former), Honors Hall Council

Feb '24 – May '25

ADHC Representative, National Collegiate Honors Council Annual Conference

Nov '24

LECTURES & FEATURES

- Lecture Series. Ethics in Science and Research, Science Mentorship Institute
- Session Moderator. Student Oral Presentation (Communications). NCUR, 2026
- Featured Presenter. 'Navigating peer review and the student journal landscape'. "Shaping Asia's Future: Development, Innovation, and Society in Transition" organized by Nazarbayev University, Asian Universities Alliance 2026
- Featured Panelist. 'It's Different for Girls' Documentary Screening and Panel. NJIT ADHC and CSLA Co-Sponsored Colloquium, NJIT 2026
- NJIT Makes: Undergraduate Research and Innovation, NJIT Promotional Material. NJIT, 2026
- Albert Dorman Honors College, Honors in Action 2025 (Research, Award, and Study Abroad Experience). ADHC Publication. NJIT, 2025
- Nayib Morán. '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum'. Article Feature. NJIT News, 2025
- Albert Dorman Honors College, Honors in Action 2024 (Research and Study Abroad Experience). ADHC Publication. NJIT, 2024
- Gurnoor Kaur. 'Undergraduate Student Researchers on Campus'. Article Feature. NJIT (Student Newspaper – The Vector), 2024
- K. Singh. 'Introduction to PyMOL and Linear Peptide Design'. Lecture. NJIT (Genetically Engineered Machines), 2024

CERTIFICATIONS & LICENSES

- New Jersey EMT License
- ACS "Stop the Bleed" Certification & Instructor License
- OSHA General Industry Safety and Health Certification
- CITI Biomedical Research Certification
- CITI RCR Basic Certification
- American Heart Association BLS & CPR for Adult/Child/Infant Certification

LEADERSHIP

Ethics Coordinator, Mentor (Neuro + Bio), Ethics Curriculum Dev (Former), Admission Officer, Symposium Committee Organizer, Grant Writer, Science Mentorship Institute (Sci-Mi)	May '24 – Present
Public Relations Chair, EMT, NJIT First Aid Squad	Oct '25 – Present, Jan '24 – Present
Stop the Bleed Instructor (Hybrid English–Punjabi), NJ Gurdwaras, NJ Office of Homeland Security and Preparedness (NJOHSP) & Sikh Youth Alliance	Jan '26 – Present
Khalsa School Virsa (Sikh History) Teacher, Sikh Sabha of New Jersey	Sep '20 – Present
Youth Camp Organizer and Coordinator, Sikh Sabha of New Jersey	Aug '21 – Present
Content Manager, The Mardana Project	May '26 – Present
Youth Camp Counselor, Guru Nanak Sikh Heritage	Aug '21 – Aug '25
President, Events Coordinator (Former), NJIT Sikhs	Dec '23 – May '26
President, Treasurer (Former), Co-Founder, NJIT MedicZero	Nov '23 – May '26
Co-Founder, Delta Epsilon Mu (Pre-medical Professional Fraternity)	Feb '24 – May '25
Fellow, Honors Summer Research Institute Fellowship	Jun '24 – Jul '24, Jun '25 – Jul '25
EMT Senior Squad Member, Cadet Crew Chief (Former), Marlboro First Aid Squad	Apr '22 – Jan '25
Fellow, No Patient Left Behind Fellowship	Jun '24 – Aug '24
Young Scholar, American College of Cardiology	May '23 – Apr '24